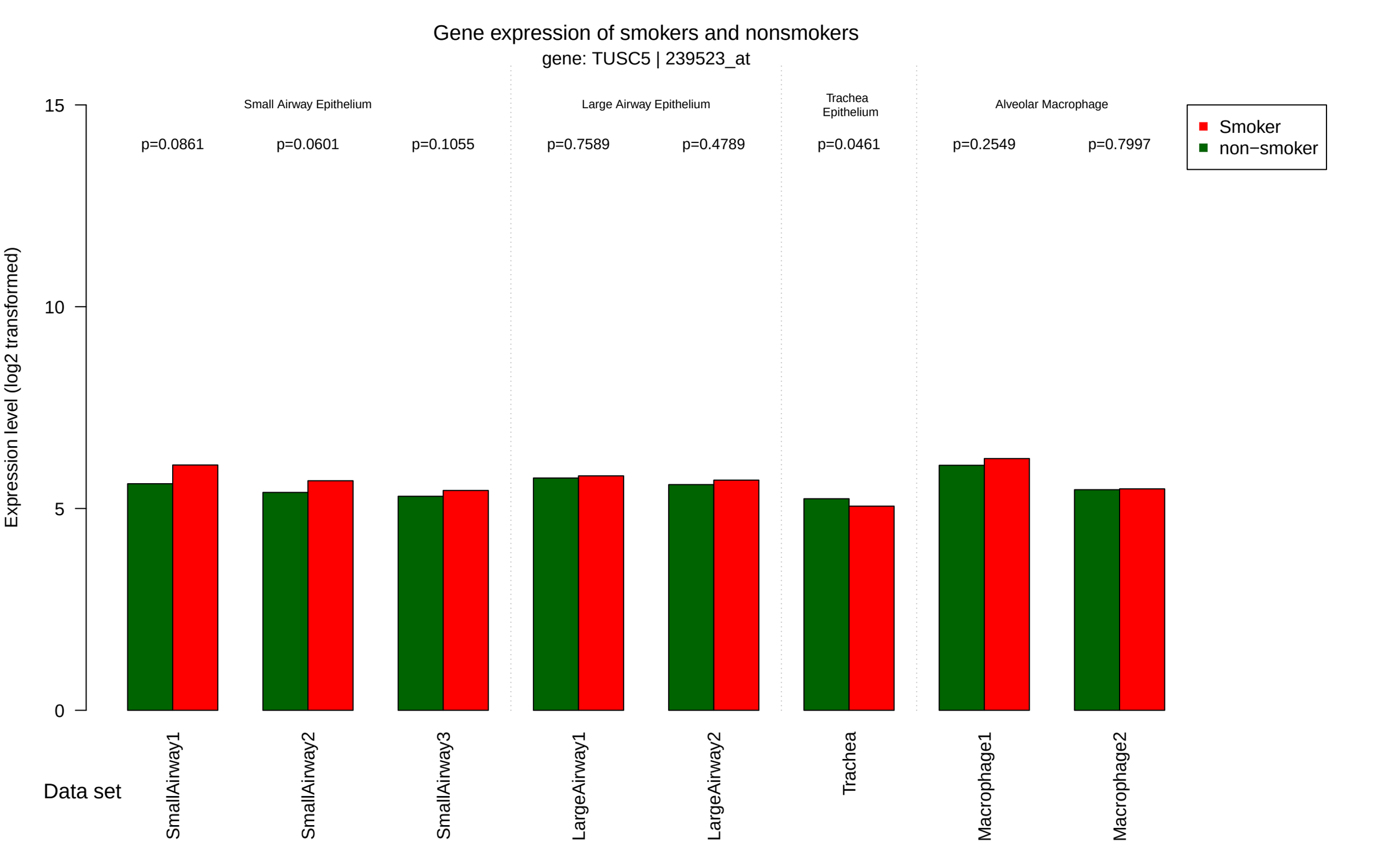
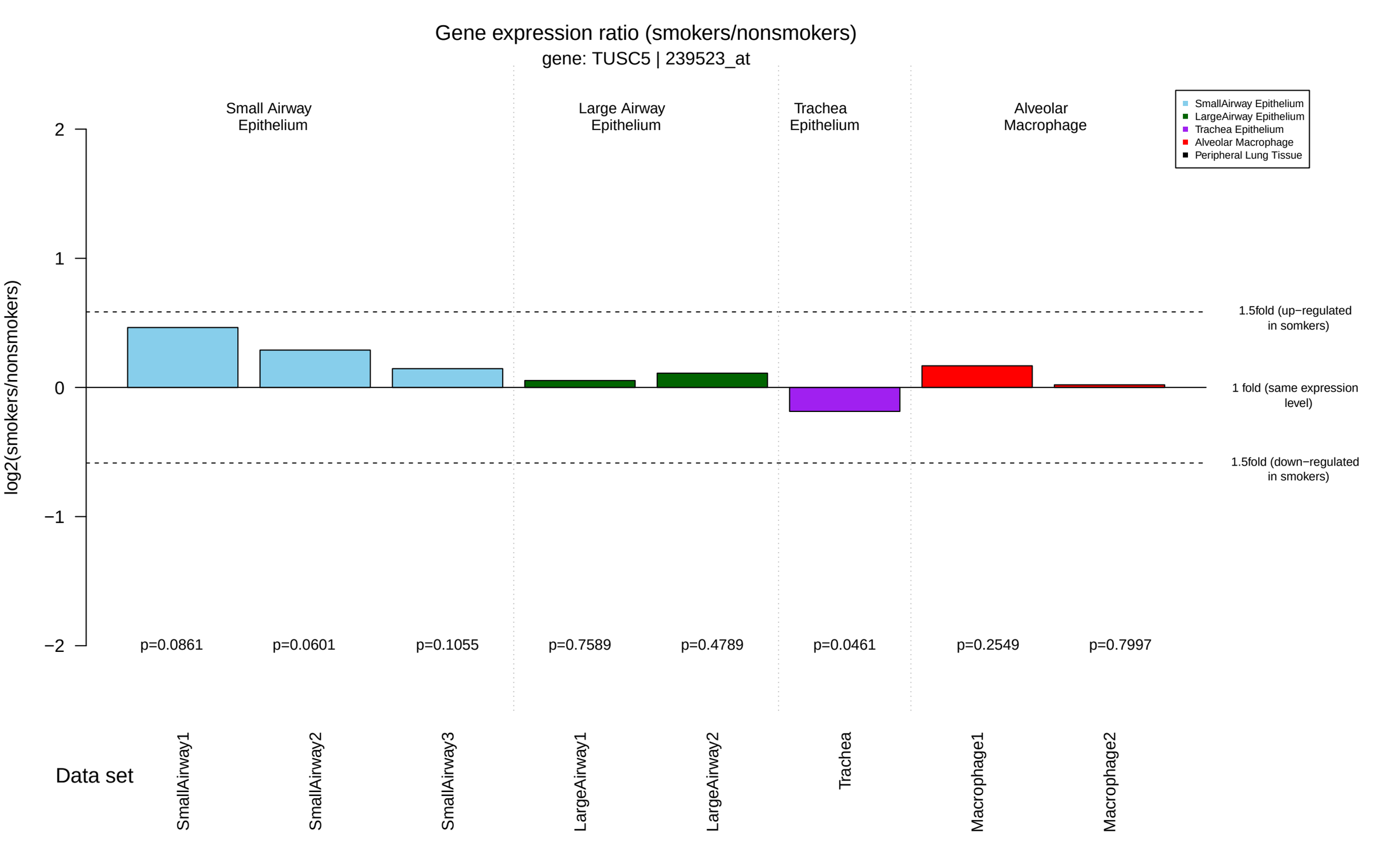
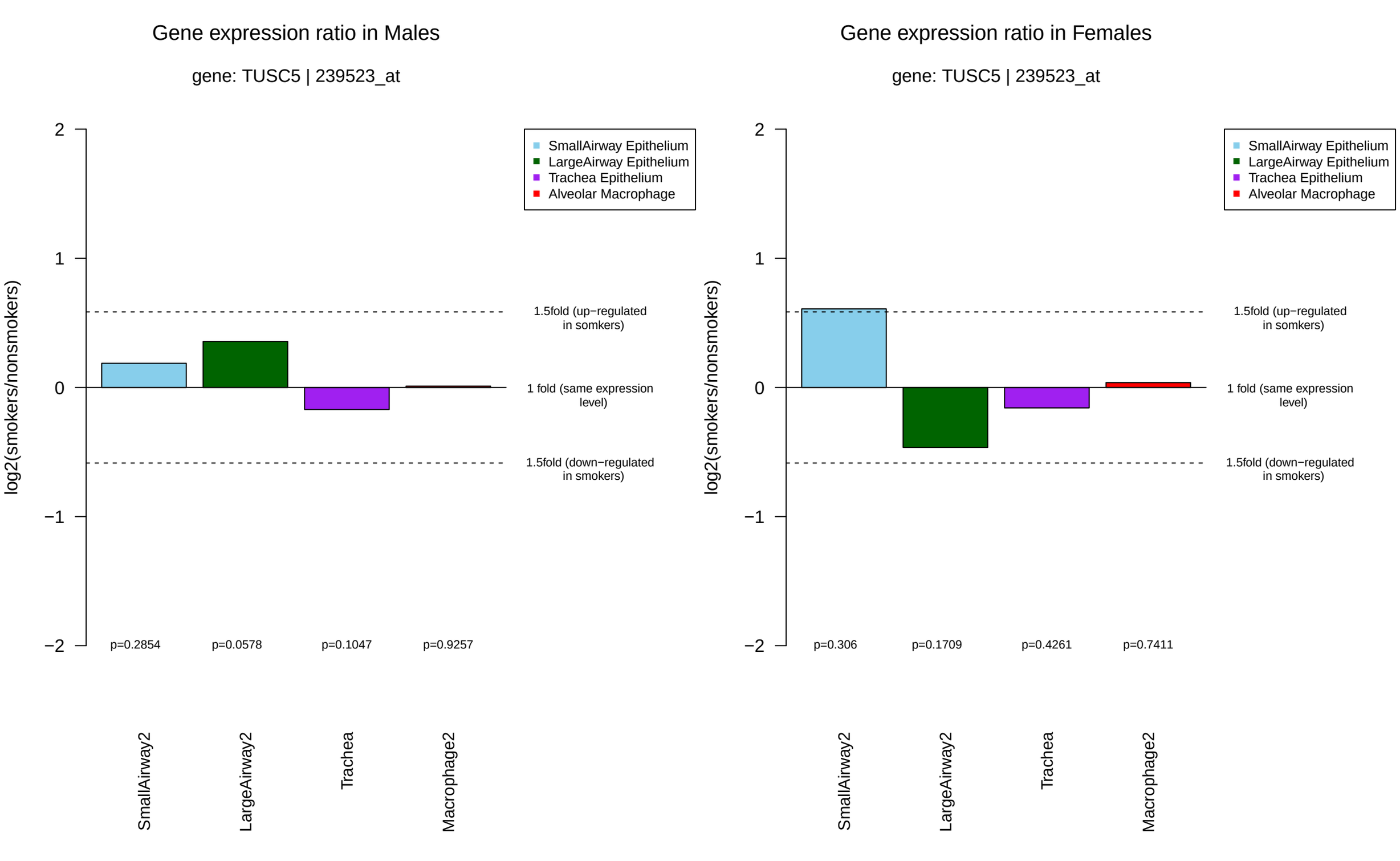
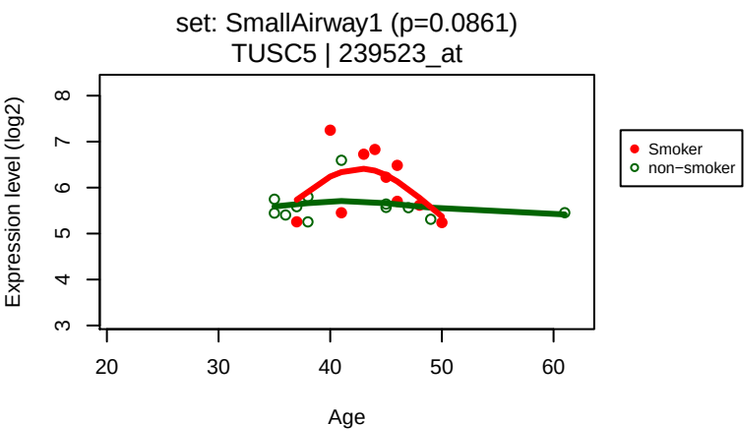
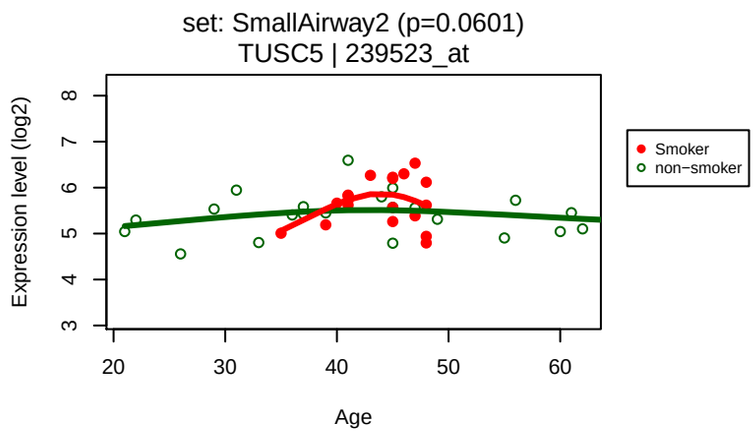
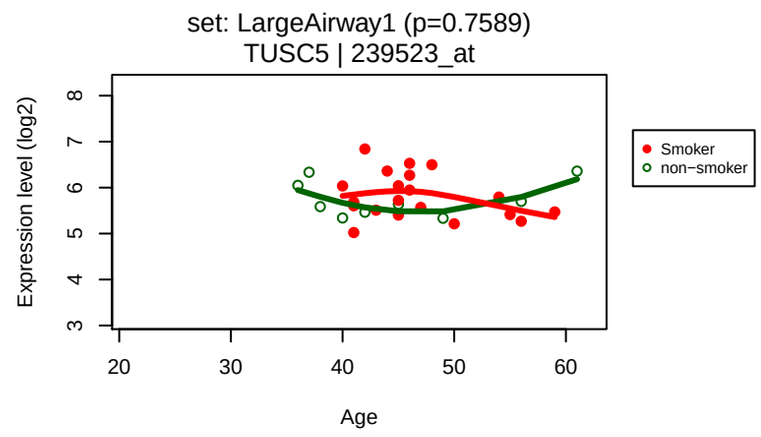

Gene expression of smokers and nonsmokers
gene: TUSC5 | 239523_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0861
p=0.0601
p=0.1055
p=0.7589
p=0.4789
p=0.0461
p=0.2549
p=0.7997
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

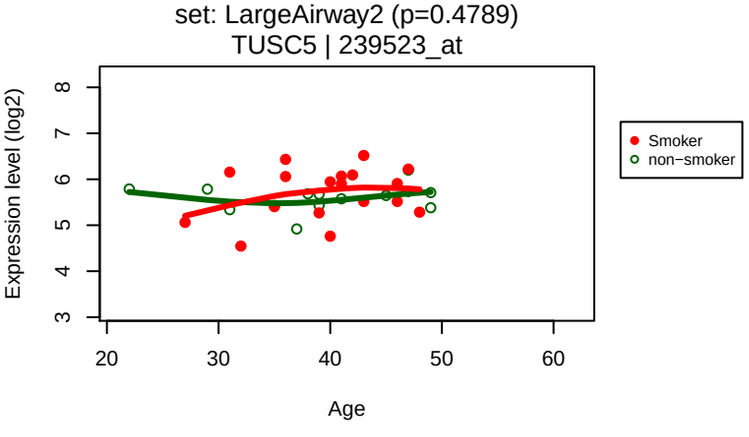

Gene expression ratio (smokers/nonsmokers)
gene: TUSC5 | 239523_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0861
p=0.0601
p=0.1055
p=0.7589
p=0.4789
p=0.0461
p=0.2549
p=0.7997
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

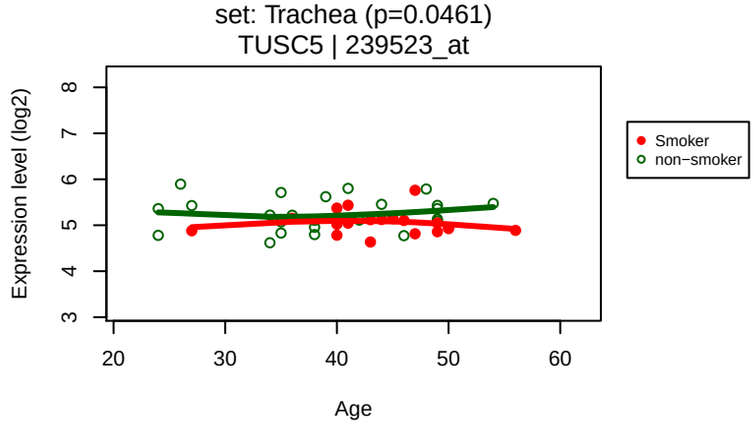

Gene expression ratio in Males
gene: TUSC5 | 239523_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2854
p=0.0578
p=0.1047
p=0.9257
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TUSC5 | 239523_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.306
p=0.1709
p=0.4261
p=0.7411
SmallAirway2
LargeAirway2
Trachea
Macrophage2

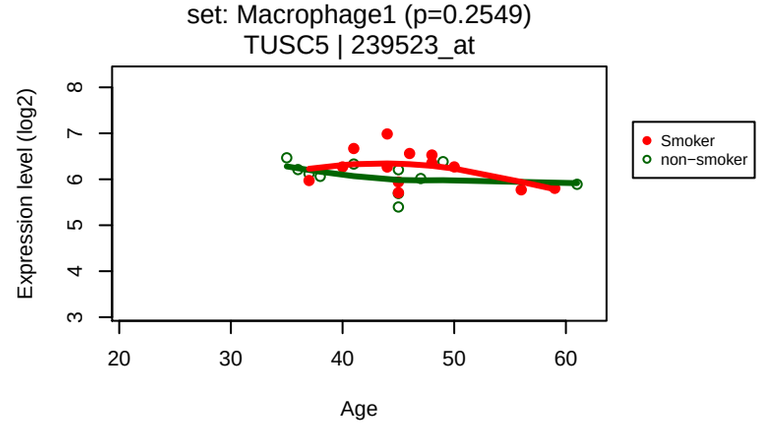

set: SmallAirway1 (p=0.0861)
TUSC5 | 239523_at
Expression level (log2)
Age
Smoker
non−smoker

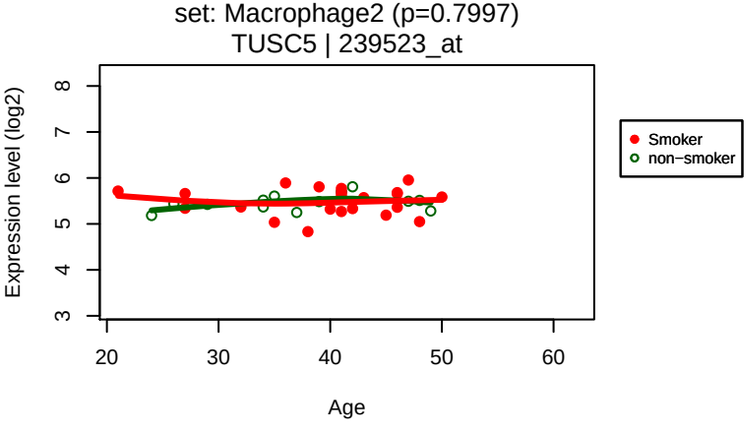

set: SmallAirway2 (p=0.0601)
TUSC5 | 239523_at
Expression level (log2)
Age
Smoker
non−smoker

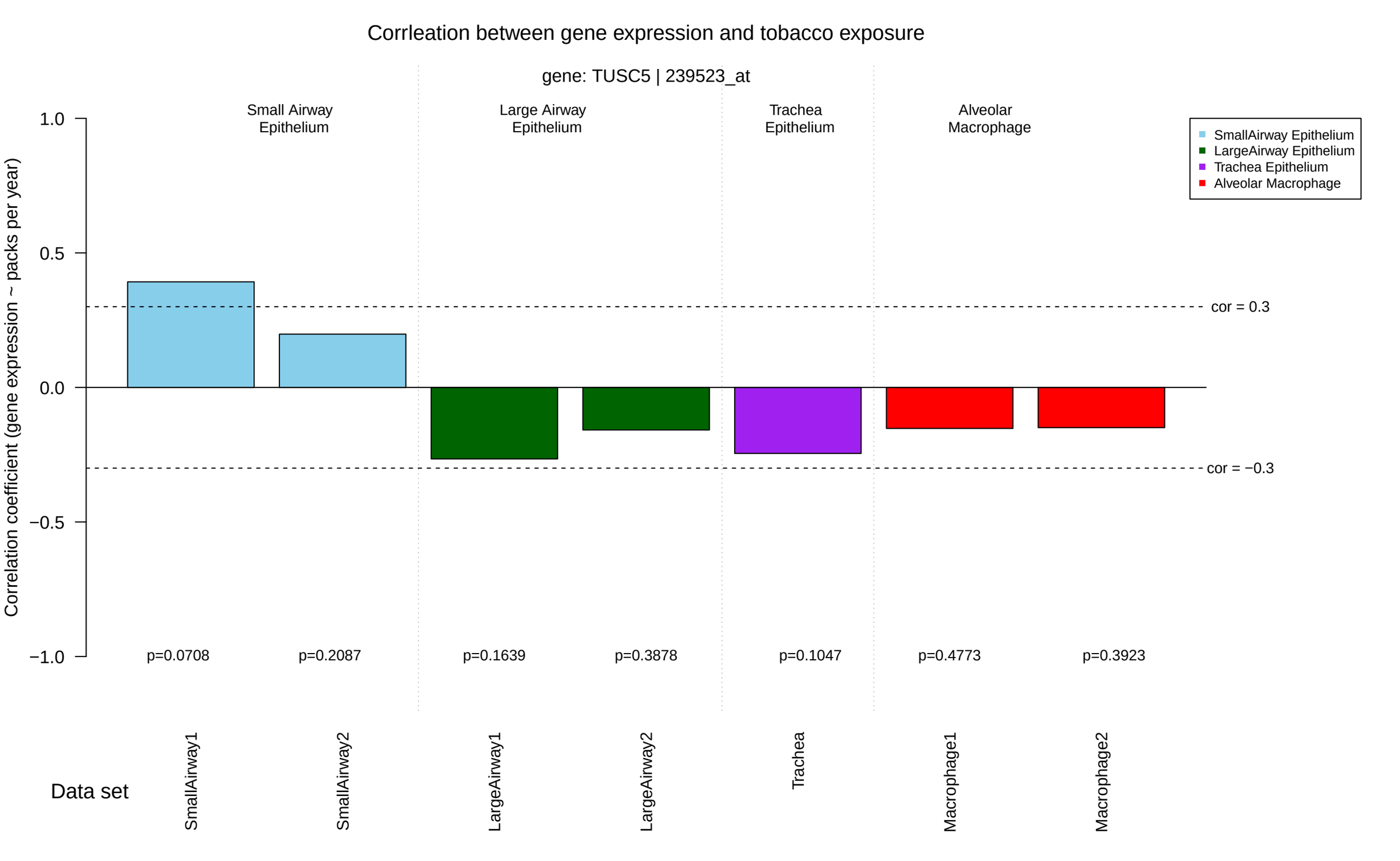

set: LargeAirway1 (p=0.7589)
TUSC5 | 239523_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4789)
TUSC5 | 239523_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0461)
TUSC5 | 239523_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2549)
TUSC5 | 239523_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7997)
TUSC5 | 239523_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TUSC5 | 239523_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0708
p=0.2087
p=0.1639
p=0.3878
p=0.1047
p=0.4773
p=0.3923
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2